

Capstone Checkpoint 2.1: Capstone Agent Design Refinement

Describe the reasoning loop your agent will use. This may be based on ReAct or a similar framework. Explain how reasoning steps guide actions and how observations influence subsequent decisions.

- The agent will follow the ReAct framework. Given a requirement, it will reason what the best step is, take action, and then evaluate/observe that action's effects. If the requirement was not satisfied, it will repeat starting with the reason step.
- An example would be to find a flight and hotel. First it will reason such as where am I flying from, where am I flying to, should it book a flight with multiple stops or direct, etc. Next, it will act by making api calls and finding all the available flights. Based on what it thought, it will filter and select the best option. It will then observe the outcome, and ask, does it solve the problem. Perhaps the only flight that meets all the user's requirements is over budget. It would then observe, learn that lesson, and either broaden its search and present the problem to the user and offer followup solutions.
- Afterwards, it would solve the hotel problem in a similar manner. However, it has an additional requirement of making sure that the hotel dates/times align with the flight details.

Identify the type of memory your agent requires. Specify whether the task relies on short-term memory, long-term memory, or both, and explain when and why that memory is needed.

- Between semantic, episodic, and procedural memory, the agent will primarily focus on semantic and episodic memory. It would have to recall previous conversations to discover what the user's preferences are.
- The agent will require both long and short term memory. It requires short term memory while conversing with the user, researching, and building the itinerary. It needs to recall the dates, the trip specific details, etc. It would also require long term to recall previous user preferences such as which seats on the plane to choose, what type of hotels, favorite attractions, etc.

Describe how your agent will use external tools. Identify at least one task in which tool usage is necessary and explain what limitation it addresses, such as computation retrieval or grounding.

- The agent will make several external tool calls such as flight, hotel, and rental car searches. It will also have to call weather api calls, searches to online forums for best attractions. By enabling and allowing the agent to act, it reduces hallucinations and knowledge gaps. It knows the most up to date attraction availability, what events are the most popular currently, the ideal weather days, live prices and flights, etc.

Explain how these additions improve the agent compared to a prompt-only approach. Reference a specific failure mode that your design resolves.

- These additions allow the agent to be far more capable. Without these, the agent would simply be a knowledgeable bot with fixed information from its training data. It wouldn't know the latest news, have knowledge of live ticket prices, or anything else along those lines. One failure mode that these additions address are hallucination and outdated information. For instance, say that model was trained up to 2014 data only, it may provide concert information with popular artists at the time who are no longer relevant today. It also won't have information of live flight tickets and may make those prices and details instead. These changes ground the model's responses and gives it access to live data.